

TARGET – E COMMERCE

SQL BUSINESS CASE STUDY

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SQL PLATFORM USED: GOOGLE BIGQUERY

PROBLEM STATEMENT:

Assuming you are a data analyst/ scientist at Target, you have been assigned the task of analysing the given dataset to extract valuable insights and provide actionable recommendations.

1. What does 'good' look like? *Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset.*

1.1. Data type of all columns in the "customers" table.

SQL Query:

```
SELECT *
FROM target_sql.customers;
```

Results

Query results							+ Create conversation	Save results
Job information	Results	Visualisation	JSON	Execution details	Execution graph			
Row	customer_id	customer_unique_id	customer_zip_co...	customer_city	customer_state			
1	2201362e68992f654942dc006...	f7d7fc0a59ef4363fdce6e3aa06...	69900	rio branco	AC			
2	31dbc13addc753e210692eaca...	5dbba6c01268a8ad43f79157bf...	69900	rio branco	AC			
3	dad907e170748a35ef4e92238...	36b1c0516f123351ffa8743041...	69900	rio branco	AC			
4	888d2ebe1af2a8c93c75dae5df...	721d1092e1a6460c67e6a0e69...	69900	rio branco	AC			
5	8a0108267d9258a0ec9f74381...	7a2dc4682890550ebe3b8befce...	69900	rio branco	AC			
6	5880e46677c68394bda62479f...	5dbba6c01268a8ad43f79157bf...	69900	rio branco	AC			
7	53996870173a3a001a1fb56ef0...	2624230437101e0bdcea1e483...	69900	rio branco	AC			
8	cd281c1a7d26cd29a3ed4b029...	086d6b5b5ba195a91aa0a6ec8...	69900	rio branco	AC			
9	717a3a9459f9006f23e24a684b...	5203034a29c3d294713bf8ccb...	69900	rio branco	AC			
10	2d618e470c95c9b425cbb0cbc...	f7d7fc0a59ef4363fdce6e3aa06...	69900	rio branco	AC			
Results per page: 50 1 - 50 of 99441								

Insights:

- We can see that the `customers` table contains 5 columns and 99441 rows.

Data Types:

- customer_id: STRING
- customer_unique_id: STRING
- customer_zip_code: INTEGER
- customer_city: STRING
- customer_state: STRING

1.2. Get the time range between which the orders were placed.

SQL Query:

- The following is for the time of orders being placed:

```
SELECT
  FORMAT_DATE('%a', order_purchase_timestamp),
  MIN(EXTRACT(TIME FROM order_purchase_timestamp)) AS earliest_time_of_order,
  MAX(EXTRACT(TIME FROM order_purchase_timestamp)) AS latest_time_of_order,
  AVG(EXTRACT(HOUR FROM order_purchase_timestamp)) AS avg_time_of_ordering
FROM target_sql.orders
WHERE order_status = 'delivered'
GROUP BY FORMAT_DATE('%a', order_purchase_timestamp);
```

- And, this one is for the dates:

```
SELECT
  MAX(DATE(order_purchase_timestamp)) AS latest_order_placed,
  MIN(DATE(order_purchase_timestamp)) AS first_order_placed,
  DATE_DIFF(MAX(DATE(order_purchase_timestamp)),
            MIN(DATE(order_purchase_timestamp)),MONTH) AS order_month_span
FROM target_sql.orders
WHERE order_status = 'delivered';
```

Results:

Query results				
Job information		Results	Visualisation	JSON
Row	f0_	earliest_time_of_...	latest_time_of_or...	avg_time_of_orde...
1	Thu	00:00:00	23:59:55	14.54122739649...
2	Mon	00:00:01	23:59:54	15.00821603719...
3	Wed	00:00:09	23:59:58	14.57468824621...
4	Tue	00:00:02	23:59:54	14.85286718699...
5	Fri	00:00:21	23:59:48	14.16587504567...
6	Sat	00:00:07	23:59:58	14.74306016106...
7	Sun	00:00:10	23:59:59	15.63429308122...

Query results			
Job information		Results	Visualisation
Row	latest_order_plac...	first_order_placed	order_month_span
1	2018-08-29	2016-09-15	23

Insights:

- From the result, we can conclude that Order activity is concentrated during afternoon hours, with average ordering time across weekdays clustering around 14:00–15:00. This analysis suggests to keep the platform working uninterrupted during this time period and handle any issues efficiently and ensure smooth working of the platform.
- The Dates spans from Sept 09 2016 to Aug 29, 2018, for customers whose orders have been delivered.

1.3. Count the Cities & States of customers who ordered during the given period

SQL Query:

```
SELECT
  COUNT(DISTINCT c.customer_state) AS count_of_states,
  COUNT(DISTINCT c.customer_city) AS count_of_cities
FROM target_sql.customers c JOIN target_sql.orders o
ON c.customer_id = o.customer_id;
```

Results:

Query results			
Job information		Results	Visualisation
Row	count_of_states	count_of_cities	
1	27	4119	

Insights:

Customers are spanned across 27 states and 4119 cities within Brazil. This analysis helps to understand the territorial reach of business firm, depth of market, helps to create strategy for further operational expansion and logistics improvement.

2. In-depth Exploration:

2.1. Is there a growing trend in the no. of orders placed over the past years?

SQL Query:

```
SELECT
    EXTRACT(YEAR FROM order_purchase_timestamp) AS years,
    COUNT(order_id) AS total_orders
FROM target_sql.orders
GROUP BY EXTRACT(YEAR FROM order_purchase_timestamp)
ORDER BY years;
```

Results:

Query results			
Job information		Results	Visualisation
Row	years	total_orders	
1	2016	329	
2	2017	45101	
3	2018	54011	

Insights:

- The No. of Orders placed has increased over the years, it indicates a growing trend in customer footfall and demand.
- Order volume shows strong year-over-year growth, indicating expanding platform adoption., hence preparation regarding logistics plan, inventory and campaigns will ensure smooth operations.

2.2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

SQL Query:

```
SELECT
    FORMAT_DATE('%B',order_purchase_timestamp) AS months,
    COUNT(order_id) AS total_orders
FROM target_sql.orders
GROUP BY FORMAT_DATE('%B',order_purchase_timestamp),
    EXTRACT(MONTH FROM order_purchase_timestamp)
ORDER BY EXTRACT(MONTH FROM order_purchase_timestamp);
```

Results:

Row	months	total_orders
1	January	8069
2	February	8508
3	March	9893
4	April	9343
5	May	10573
6	June	9412
7	July	10318
8	August	10843
9	September	4305
10	October	4959
11	November	7544
12	December	5674

Insights:

- Clear seasonal trend with initial increase then gradual decrease, and fluctuating demand. The peak months are May, July and August with more than 10K orders. These months indicate customer demand is high, reasons could be festivals, sales, events, etc, during these periods inventory and logistics need to be handled efficiently.
- In the last quarter of year, the demand is low, these can be targeted with marketing campaigns, promotions and discounts.

2.3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

SQL Query:

```
WITH hours AS (  
    SELECT order_id,  
    EXTRACT(HOUR FROM order_purchase_timestamp) as oh  
    FROM target_sql.orders  
) ,  
tod AS (  
    SELECT  
        order_id,  
        CASE  
            WHEN oh BETWEEN 0 AND 6 THEN 'Dawn'  
            WHEN oh BETWEEN 7 AND 12 THEN 'Morning'  
            WHEN oh BETWEEN 13 AND 18 THEN 'Afternoon'  
            ELSE 'Night' END AS time_of_day  
    FROM hours  
)  
SELECT  
    time_of_day,  
    COUNT(order_id) AS total_orders  
FROM tod  
GROUP BY time_of_day  
ORDER BY  
CASE WHEN time_of_day = 'Dawn' THEN 1  
    WHEN time_of_day = 'Morning' THEN 2  
    WHEN time_of_day = 'Afternoon' THEN 3  
    ELSE 4 END;
```

Results:

Row	time_of_day	total_orders
1	Dawn	5242
2	Morning	27733
3	Afternoon	38135
4	Night	28331

Insights:

- Brazilians Order mostly during afternoon period, which indicates their activity behaviour. During dawn there has been less activity, and regular during remaining period.
- This analysis helps to understand server traffic and ensure server handling capability during peak hours, logistics preparedness and, use of marketing ads.

3. Evolution of E-commerce orders in the Brazil region

3.1. Get the month on month no. of orders placed in each state.

SQL Query:

```
SELECT
    c.customer_state,
    FORMAT_DATETIME('%Y-%m', o.order_purchase_timestamp) AS order_months,
    COUNT(o.order_id) AS total_orders
FROM target_sql.customers c JOIN target_sql.orders o
    ON c.customer_id = o.customer_id
GROUP BY
    c.customer_state,
    FORMAT_DATETIME('%Y-%m', o.order_purchase_timestamp)
ORDER BY c.customer_state, order_months;
```

Results:

Row	customer_state	order_months	total_orders
1	AC	2017-01	2
2	AC	2017-02	3
3	AC	2017-03	2
4	AC	2017-04	5
5	AC	2017-05	8
6	AC	2017-06	4
7	AC	2017-07	5
8	AC	2017-08	4
9	AC	2017-09	5
10	AC	2017-10	6

(565 Rows)

Insights:

- No. of Orders placed across states and months varies significantly; fluctuations are obvious from the analysis. The higher no. of orders across months in a state signifies the higher customer demands, more population and higher reach of business. These states have been performing efficiently.
- States such as RJ and SP consistently rank among top contributors in monthly order volume.
- There are few states which are under performing and even do not have no. of order in double digit. Such as State RR. Such states indicate low demand, less reach and indicate less operations. They offer opportunities for market expansion and operations setup, which can be achieved by identifying geographical condition, customer awareness about e commerce awareness.

3.2. How are the customers distributed across all the states?

SQL Query:

```
SELECT
```

```
customer_state,
COUNT(customer_unique_id) as total_customers
FROM target_sql.customers
GROUP BY customer_state
ORDER BY total_customers DESC;
```

Results:

Row	customer_state	total_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

(27

Rows)

Insights:

- SP, RJ, and MG account for the highest customer counts, indicating stronger market penetration in these regions. All other states have no. of customer less than half of average of above three states. These states are underrepresented and have low customers.
- With this analysis business firm can target low customer states for market expansion; focus on inventory, logistics in high customer states and run region specific campaigns for more reach.
- **From previous analysis 3.1 and this 3.2, it is evident that, it is not necessary more no. of customer in state should create more no. of orders. Hence region specific, depth of market, ease of operations, marketing creates affects business.**

4. *Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.*

4.1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

SQL Query:

```
WITH tcs AS (
  SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
    ROUND(SUM(p.payment_value),2) AS order_total_cost
  FROM target_sql.orders o JOIN target_sql.payments p
    ON o.order_id = p.order_id
  WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017,2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 01 AND 08
  GROUP BY EXTRACT(YEAR FROM o.order_purchase_timestamp)
), lags AS (
  SELECT
    order_year,
    order_total_cost,
    LAG(order_total_cost) OVER (
      ORDER BY order_year
    ) AS pre_year_cost
  FROM tcs
)
SELECT
  order_year,
  order_total_cost,
  pre_year_cost,
  ROUND((order_total_cost - pre_year_cost)/pre_year_cost * 100,2) AS
percentage_growth
FROM lags
ORDER BY order_year;
```

Results:

Row	order_year	order_total_cost	pre_year_cost	percentage_growth
1	2017	3669022.12	null	null
2	2018	8694733.84	3669022.12	136.98

Insights:

- This analysis shows significant growth in revenue in 2018 as compared to 2017. The total order cost increased by 137% from 2017 to 2018.
- It depicts the plans, campaigns, investments decisions have brought profit to the firm. Customer demand is increased, which call for e-commerce operations expansion.

4.2. Calculate the Total & Average value of order price for each state.

SQL Query:

The problem can be solved using two different approaches:

Approach 1 using only State level aggregation

```
SELECT
    c.customer_state,
    ROUND(SUM(i.price),2) AS order_total_value,
    ROUND(AVG(i.price),2) AS order_avg_value
FROM target_sql.orders o
    JOIN target_sql.order_items i ON i.order_id = o.order_id
    JOIN target_sql.customers c ON c.customer_id = o.customer_id
GROUP BY c.customer_state;
```

Approach 2 using Order Level totals, and then State level aggregation:

```
WITH cte1 AS (
    SELECT
        i.order_id,
        c.customer_state,
        SUM(i.price) AS order_total_price
    FROM target_sql.order_items i
        JOIN target_sql.orders o ON o.order_id = i.order_id
        JOIN target_sql.customers c ON c.customer_id = o.customer_id
    GROUP BY i.order_id, c.customer_state
)
SELECT
    customer_state,
    SUM(order_total_price) AS total_order_value,
    ROUND(AVG(order_total_price),2) AS avg_order_value
FROM cte1
GROUP BY customer_state;
```

Results:

Row	customer_state	order_total_value	order_avg_value
1	MG	1585308.03	120.75
2	SP	5202955.05	109.65
3	RS	750304.02	120.34
4	RJ	1824092.67	125.12
5	SC	520553.34	124.65
6	PR	683083.76	119.0
7	ES	275037.31	121.91
8	GO	294591.95	126.27
9	DF	302603.94	125.77
10	MA	119648.22	145.2

Approach 1 (27 Rows)

Row	customer_state	total_order_value	avg_order_value
1	SP	5202955.049999...	125.75
2	RJ	1824092.669999...	142.93
3	PR	683083.760000...	136.67
4	RS	750304.020000...	138.13
5	MG	1585308.030000...	137.33
6	GO	294591.949999...	146.78
7	SC	520553.339999...	144.12
8	PB	115268.080000...	216.67
9	DF	302603.939999...	142.4
10	PA	178947.810000...	184.48

Approach 2 (27 rows)

Insights:

- From either of the analysis, It is evident about high variation in average order values across states. This indicates differences in purchasing types and behaviours, customer spending behaviours. It will help in offering services to high value states for retaining customers.
- Strategies can implemented to improve operations, logistics, marketing in particular regions. Identify KPI's in high value regions and understand why less value in low value states.
- Approach 2 provides a more accurate representation of average order value by first aggregating item prices at order level before state-level averaging.

4.3. Calculate the Total & Average value of order freight for each state.

SQL Query:

```
WITH cte1 AS (  
  SELECT  
    order_id,  
    c.customer_state,  
    SUM(freight_value) AS total_freight_value  
  FROM target_sql.orders o  
    JOIN target_sql.order_items i USING (order_id)  
    JOIN target_sql.customers c USING (customer_id)  
  GROUP BY order_id, c.customer_state  
)  
SELECT  
  customer_state,  
  ROUND(SUM(total_freight_value),2) AS total_order_freight_value,  
  ROUND(AVG(total_freight_value),2) AS avg_freight_value  
FROM cte1  
GROUP BY customer_state  
ORDER BY total_order_freight_value DESC;
```

Results:

Row	customer_state	total_order_freight_value	avg_freight_value
1	SP	718723.07	17.37
2	RJ	305589.31	23.95
3	MG	270853.46	23.46
4	RS	135522.74	24.95
5	PR	117851.68	23.58
6	BA	100156.68	29.83
7	SC	89660.26	24.82
8	PE	59449.66	36.07
9	GO	53114.98	26.46
10	DF	50625.5	23.82

(27 rows)

Insights:

- Freight Value varies for each state. Some states incur higher total and average freight values, which indicates increase in logistics costs. Such states need local warehouses setup, better logistics routes and plans, local sellers to be introduced.

5. Analysis based on sales, freight and delivery time.

5.1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

SQL Query:

```
SELECT
    order_id,
    DATE(order_purchase_timestamp) AS order_placed_date,
    DATE(order_delivered_customer_date) AS order_delivered_cust_date,
    DATE(order_estimated_delivery_date) AS estimated_delivery_date,
    DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS
time_to_deliver,
    DATE_DIFF(order_delivered_customer_date, order_estimated_delivery_date, DAY)
AS diff_estimated_delivery
FROM target_sql.orders
WHERE order_delivered_customer_date IS NOT NULL;
```

Results:

Row	order_id	order_plac...	order_delic...	estimated...	time_to_d...	diff_estimated_delivery
1	770d331c84e5b214bd9dc70a10b829d0	2016-10-07	2016-10-14	2016-11-29	7	-45
2	dabf2b0e35b423f94618bf965fcb7514	2016-10-09	2016-10-16	2016-11-30	7	-44
3	8beb59392e21af5eb9547ae1a9938d06	2016-10-08	2016-10-19	2016-11-30	10	-41
4	1950d777989f6a877539f53795b4c3c3	2018-02-19	2018-03-21	2018-03-09	30	12
5	bfb0f9bdef84302105ad712db648a6c	2016-09-15	2016-11-09	2016-10-04	54	36
6	cd3b8574c82b42fc8129f6d502690c3e	2016-10-03	2016-10-14	2016-11-23	10	-39
7	31b0dd6152d2e471443debf037ae171d	2016-10-05	2016-10-13	2016-11-23	8	-40
8	a8c3f65a43b2e956cbfca10db4300853	2016-10-04	2016-10-20	2016-11-24	15	-34
9	a041155864e5141116458291313b35c2	2016-10-04	2016-10-13	2016-11-24	9	-41
10	50013835d7b14aefb452825864d3e414	2016-10-04	2016-10-21	2016-11-24	17	-33

(96476 rows)

Insights:

- Delivery times vary materially, with both early and delayed deliveries observed, indicating inconsistency in logistics performance. This indicates logistics performance showcase, inventory fulfilments.
- Positive delivery differences indicate delayed deliveries, whereas Negative delivery differences indicate early deliveries.
- Delayed deliveries may negatively impact customer goodwill, while early deliveries enhance customer experience and trust, retention.

5.2. Find out the top 5 states with the highest & lowest average freight value.

SQL Query:

```
WITH cte1 AS (  
    SELECT  
        order_id,  
        c.customer_state,  
        ROUND(SUM(i.freight_value),2) AS order_freight_value  
    FROM target_sql.orders o  
        JOIN target_sql.order_items i USING (order_id)  
        JOIN target_sql.customers c USING (customer_id)  
    GROUP BY order_id, customer_state  
,  
cte2 AS (  
    SELECT  
        customer_state,  
        ROUND(AVG(order_freight_value),2) AS total_avg_freight_value,  
        DENSE_RANK() OVER (  
            ORDER BY AVG(order_freight_value) DESC) AS highranks,  
        DENSE_RANK() OVER (  
            ORDER BY AVG(order_freight_value)) AS lowranks  
    FROM cte1  
    GROUP BY customer_state  
    ORDER BY total_avg_freight_value DESC  
)  
SELECT  
    customer_state,  
    total_avg_freight_value,  
    CASE  
        WHEN highranks <= 5 THEN 'top 5 highest'  
        WHEN lowranks <=5 THEN 'top 5 lowest'  
    END AS top5_high_low  
FROM cte2  
WHERE highranks <= 5 OR lowranks <=5  
ORDER BY total_avg_freight_value DESC;
```

Results:

Row	customer_state	total_avg_freight_value	top5_high_low
1	RR	48.59	top 5 highest
2	PB	48.35	top 5 highest
3	RO	46.22	top 5 highest
4	AC	45.52	top 5 highest
5	PI	43.04	top 5 highest
6	RJ	23.95	top 5 lowest
7	DF	23.82	top 5 lowest
8	PR	23.58	top 5 lowest
9	MG	23.46	top 5 lowest
10	SP	17.37	top 5 lowest

Insights:

- Average Freight Value differs for each state, highest being for RR state with 48.59. The top 5 states with highest average freight value indicates warehouse distance relevance. Such states call for warehouse expansion and more seller onboarding, which will lower Freight charges and customer satisfaction.
- Average freight charges are low for state SP with 17.37. The Top 5 states with lowest freight charges indicate warehouse, seller quick access for logistics operation. They attract opportunity for scaling business and market density.

5.3. Find out the top 5 states with the highest & lowest average delivery time.

SQL Query:

```
WITH dt AS (  
SELECT  
    o.order_id,  
    c.customer_state,  
    DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS  
        time_to_deliver  
FROM target_sql.orders o  
    JOIN target_sql.customers c USING (customer_id)  
WHERE order_delivered_customer_date IS NOT NULL  
)  
tr AS (  
SELECT  
    customer_state,  
    ROUND(AVG(time_to_deliver),2) AS avg_delivery_time,  
    DENSE_RANK() OVER ( ORDER BY AVG(time_to_deliver) DESC ) AS high_rank,  
    DENSE_RANK() OVER ( ORDER BY AVG(time_to_deliver) ) AS low_rank  
FROM dt  
GROUP BY customer_state  
)  
SELECT
```

```

customer_state,
avg_delivery_time,
CASE
    WHEN high_rank <= 5 THEN 'top 5 highest'
    WHEN low_rank <=5 THEN 'top 5 lowest'
END AS top5_high_low
FROM tr
WHERE high_rank <=5 OR low_rank <= 5
ORDER BY avg_delivery_time DESC;

```

Results:

Row	customer_state	avg_delivery_time	top5_high_low
1	RR	28.98	top 5 highest
2	AP	26.73	top 5 highest
3	AM	25.99	top 5 highest
4	AL	24.04	top 5 highest
5	PA	23.32	top 5 highest
6	SC	14.48	top 5 lowest
7	DF	12.51	top 5 lowest
8	MG	11.54	top 5 lowest
9	PR	11.53	top 5 lowest
10	SP	8.3	top 5 lowest

Insights:

- Some states incur significantly higher average delivery times relative to lowest average time of delivery. The RR state experiences highest average delivery time of 29 days, RR's delivery time is ~3.5x higher than SP, highlighting significant regional logistics disparity.
- This significant difference shows varying logistics performance pan Brazil. This will downgrade the reputation of the firm. To maintain average delivery time similar for all states, the states with higher delivery times need improvement in logistics infrastructure, warehouse placement in major cities, optimized routes for earliest delivery.

5.4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

SQL Query

```

WITH dd AS (
SELECT
    o.order_id,
    c.customer_state,
    DATE_DIFF(order_delivered_customer_date,order_estimated_delivery_date,
              DAY) AS del_days_diff
FROM target_sql.customers c
    JOIN target_sql.orders o USING (customer_id)
WHERE order_delivered_customer_date IS NOT NULL

```

```

), ddr AS (
SELECT
    customer_state,
    ROUND(AVG(del_days_diff),2) AS avg_del_days,
    DENSE_RANK() OVER ( ORDER BY AVG(del_days_diff) ) AS del_ranks
FROM dd
GROUP BY customer_state
)
SELECT
    customer_state,
    avg_del_days,
    del_ranks AS top_5_fastest_delivery
FROM ddr
WHERE del_ranks <= 5
ORDER BY avg_del_days;

```

Results:

Row	customer_state	avg_del_days	top_5_fastest_delivery
1	AC	-19.76	1
2	RO	-19.13	2
3	AP	-18.73	3
4	AM	-18.61	4
5	RR	-16.41	5

Insights:

- These states consistently perform better than all others in delivery expectations, with orders being delivered much before than estimated time (-ve sign indicates early delivery. This indicates outstanding logistics performance, warehouse placements, and consistent communication networks.
- For the underperforming states, these states performance can be used as operational benchmarks and set a margin.

6. Analysis based on the payments:

6.1. Find the month on month no. of orders placed using different payment types.

SQL Query:

```
SELECT
  FORMAT_DATETIME('%Y-%m', o.order_purchase_timestamp) AS order_month,
  p.payment_type,
  COUNT(DISTINCT p.order_id) AS total_orders
FROM target_sql.payments p
  JOIN target_sql.orders o USING (order_id)
GROUP BY
  FORMAT_DATETIME('%Y-%m', o.order_purchase_timestamp),
  p.payment_type
ORDER BY order_month, p.payment_type;
```

Results:

Row	order_month	payment_type	total_orders
1	2016-09	credit_card	3
2	2016-10	UPI	63
3	2016-10	credit_card	253
4	2016-10	debit_card	2
5	2016-10	voucher	11
6	2016-12	credit_card	1
7	2017-01	UPI	197
8	2017-01	credit_card	582
9	2017-01	debit_card	9
10	2017-01	voucher	33

(90 rows)

Insights:

- Credit Card and UPI payment have been used mostly by customer for payments. There has been a varying trends across months. Vouchers are being used mostly during year end or start, which indicates the gifting options, collection and usage of vouchers for benefits.
- Understanding payment preferences helps in:
Optimizing payment options, Offering targeted discounts and Improving customer experience.

6.2. Find the no. of orders placed on the basis of the payment installments that have been paid.

SQL Query:

```
SELECT
    payment_installments,
    COUNT(DISTINCT order_id) AS total_orders
FROM target_sql.payments
GROUP BY payment_installments
ORDER BY total_orders DESC, payment_installments;
```

Results:

Row	payment_installments	total_orders
1	1	49060
2	2	12389
3	3	10443
4	4	7088
5	10	5315
6	5	5234
7	8	4253
8	6	3916
9	7	1623
10	9	644

Insights:

- Single-installment payments dominate transaction volume, though multi-installment purchases remain significant. However, installment based payments are significant among customer, upto 10 installments.
- No cost EMI options with benefits and discounts can be promoted for customer goodwill. Affordability options can be optimized for conversion rates.

FINAL SUMMARY:

Overall Insights on Target E-commerce Business:

- Strong and consistent business growth
- Regional disparities in customer and order distribution
- Logistics efficiency varies significantly
- Payment behavior indicates demand for flexibility

Strategic Incursions:

- Optimize logistics infrastructure in high-delivery-time and high-freight states.
- Expand seller/warehouse network in underperforming regions to improve reach.
- Leverage high-performing states as operational benchmarks.
- Promote flexible installment/payment options to improve conversion.
- Prepare inventory and logistics capacity for seasonal demand peaks.